



République algérienne démocratique et populaire

Ministère de l'Enseignement Supérieur et de la Recherche Scientifique
Université de Hassiba Ben Bouali Chlef
Faculté des Sciences Exactes et de l'Informatique
Département informatique



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TP SHEET N°02

Intended for 2nd year computer science license students

Group 01 /02/03/04/05/06/07 et 08

MODULE: Information system

Date of evaluation of the TP 02 sheet November 24/25/and 26, 2024.

EXERCICES (flow diagram)

first part

Establish and design the information flow diagram on **EDRAWMAX** in the following exercises:

Exercise 01: (Justification of an absence)

A student is allowed to miss their sessions at the university if their health does not allow them to attend.

However, he must justify his absence. Within 48 hours, the student submits a request and medical documents (prescription, medical certificate) to the secretariat of their department.

The secretaries check the documents submitted. If they present a problem (processing not followed, deadline exceeded), the request is rejected and the documents are returned to the student. Otherwise, the secretaries check the student's timetable and prepare the list of sessions to be justified (tutorial and practical work sessions).

On this, proof of absence is prepared and sent to the head of department for validation and signature. After recovery, the proof is delivered to the student who presents it if requested by teachers

Exercise 02: (University scholarship registration)

At the beginning of each year, students wishing to benefit from the university scholarship must register. The student submits the requested file to the registration office at the University of Chlef (secretariat of his dean). The agent checks the documents presented (list and validity).

If a document is missing, the file is returned to the student on site. The accepted files are subsequently sent to the university works headquarters where a conformity check is carried out. If any pieces were found to be invalid, a list of those pieces is sent to the student to bring back new copies.

After submitting these documents and completing the file, a letter confirming registration with the university scholarship service is sent to the student.

Exercice 3 : (Receive note)

To retrieve a copy of the transcript, the student goes to the secretariat of their department and submits a request which mentions the year of study, the academic year and the type of transcript they wish to retrieve.

The secretary checks the department's archives.

If no problems are present, she prepares the statement and sends it to the department head for signature. The statement is given to the student after collection.

If a problem is detected (a data error, file not found, etc.), the request is given to the student with the reason for refusal.

Exercise 04: (Payment to the public treasury)

To make payments to public institutions, you must go through the public treasury of the wilaya. The administration concerned by the payment must provide the citizen with a payment voucher which contains information about this administration, the reason for the operation and the amount to be paid.

The citizen must go to the Wilaya public treasury reception desk.

The employee prepares a payment receipt in two copies based on the payment voucher.

The citizen then goes to the cash register and makes the necessary payment.

Thereupon, the cashier signs and stamps both copies; he keeps one copy and returns the other to the citizen as proof of payment to be given to the administration concerned.

Exercise 05: Study area “Following up on negligent customers”

The Faille company works with more than 150 account clients. At the start of the month, the “Accounting – Finance” department requests a report on negligent customers (customers more than 5 days late in paying an invoice) from the “Accounting customers” department. For each of these customers, a request for information is sent to the sales department via internal messaging. Based on the information collected, the service contacts customers to inform them of the delay and obtain an explanation. If the customer's financial situation does not appear too critical, she puts in place new payment conditions. Otherwise, it transmits the file to the litigation department which will initiate legal proceedings for debt recovery.

Exercise 06: Area of study “Reimbursement of travel expenses”

The procedure for managing a company's travel expenses leaves something to be desired, you are responsible for proposing improvements to the existing system.

Here is the description of the current procedure: Traveling salespeople fill out a form each month where they note the conditions of the trips made. They attach all the supporting documents (transport tickets, hotel bills, taxi bills, etc.) and submit the completed document to the accounting department. The accounting department asks the head of the sales department to validate the information provided by its employees. Once this validation has been obtained, the Accounting department enters the information noted on the form by calculating the amounts due from the supporting documents. A report is produced and sent to the sales representative for verification and signature. Once the verification has been carried out, the statements are transmitted to the financial department which makes the payment.

Second part

- Exercise 07: Organization chart

A company has the following services and managers:

- 1/Applied research office – 2/Factory director -3/ Prospecting and advertising department – 4/Personnel training department -5/ Customer accounting department -6/Management secretariat – 7/Material reception department 1st- 8/Sales director - 9/Payroll department – 10/Treasury department -11/Purchasing manager -12/President and CEO – 13/Scheduling department -14/Production workshop -15/After-sales service - 16/ Accounts payable department -17/ Accounting and financial director -18/ Methods department – 19/Personnel director.

Question: knowing that the company has 6 functions:

1-Purchasing/ 2-Production/ 3-Sales /4-Accounting and finance/ 5-Human resources management (HRM or Social)/6-Director.

Starting from the list:

Q1: Determine the functions to which each manager or each department belongs?

Q2: Trace and draw the company's organization chart on EdrawMax?

Q3: Determine its structure (type of organization chart)

Exercise 08: Organization chart & Company

A company that assembles and markets computers has a computer assembly workshop, a sales department, an accounting department and a director who is at the head of the company.

For its smooth operation, the company purchases assembly parts from different suppliers, and sells the assembled computers to different customers. The sales department is responsible for orders, reporting breakdowns and deliveries. The accounting department is responsible for invoices and payments.

At the end of each month, the sales department must provide management with sales and breakdown statistics. The accounting department must provide information about the financial status of the company. For these purposes, each of these services has software. No computer

network is installed at the company. The two software programs each run on a computer connected to a printer. In each department, there is an agent who is responsible for entering input information and editing output states.

Management closely monitors the operation of the company, and issues orders to increase or decrease production according to the information provided by the sales department and the financial department.

Questions

1. Give the organization chart of this company.
2. Classify the elements of this enterprise into decision, information and operating systems.
3. Is the information system automated? If so, what are its elements?
4. Who are the elements of this company's environment?
5. Schematize the interactions (information flow) between the information system, the decision-making system, the operating system of the company.

Exercise 09 Organization chart & Company

A manufacturing company with 1,200 employees pays these employees by check. At the end of each month, human resources staff review how many employees should be paid, fill out and cut 1,200 completed checks; one for each employee. To collect their salary, employees should go to the human resources office with their employee ID cards.

Each employee presents their identification card to the human resources staff so that they can ensure that the check was given to the correct employee.

One day, a misguided employee goes to the human resources office with a fake ID and gets the check that doesn't belong to him. As a result of the stolen check, the company experiences difficulty compensating the employee whose check was stolen. Reacting to this event, the human resources director plans to install an automated system that automates the payment process at the end of the month. Employee bank account information will be stored in the system and their pay will be deposited directly into their bank accounts at the end of each month.

Answer the following questions:

1. What is the system described in the text?
2. Who are the users of this system?
3. Give the process steps of the main activity of this system?
4. What are the activities (and their frequencies), data and controls in this process?
5. What is the objective of the proposed new system?
6. What are the conditions for the success of the new system?

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#Notice_to_students_Licence2_computer science**TP sheet N°02**

- *Date of evaluation of the TP 02 sheet will be made on November 24/25/and 26, 2024.*